04 Systems Architect: AURA Dialogic Engine

Station role: Systems Architect

Runs both dialogue modes. Pure standard library, Python 3.9+.

```
04_Systems_Architect\python run_tests.py
# 5 tests (owned folders 02 and 03)
python 04_dialogic_engine.py --mode both
# offline scripted run
python 04_dialogic_engine.py --mode both --live
# real Gemini / OpenAI / Claude (needs keys)
```

Traces are written into `02_Protocol_Officer/traces/` and `03_Security_Sentinel/traces/`.

Files in this folder

File	Purpose
`04_dialogic_engine.py`	Entry point: wires the Socratic (02) and Galilean (03) engines to the LLM back-ends
`backends.py`	Gemini, Open AI (ChatGPT), Claude adapters and the offline mock backend
`dialogue_trace.py`	Records each dialogue step; writes .md and .json traces
`run_tests.py`	Runs the tests in folders 02 and 03
`.vscode/launch.json`	VS Code run/debug configurations
`.env.example`	Template for API keys (`GEMINI_API_KEY`, `OPENAI_API_KEY`, `ANTHROPIC_API_KEY`)
`.gitignore`	Keeps `.env` and caches out of Git

Keep the four role folders side by side: the engine imports code from folders 02 and 03.